

LESSON GUIDE

Technology Plus

Unit 7: Security

6.1.8 – Non-Repudiation

Lesson Overview:

In this lesson, students will explore non-repudiation and how it relates to the AAA Triad.

Students will:

- Explain non-repudiation and explain its role in cybersecurity.
- Describe the relationship between non-repudiation and the AAA Triad.
- Apply non-repudiation concepts to real-world scenarios

Guiding Question: How can organizations ensure that users cannot deny their actions in a digital system?

Suggested Grade Levels: 8-12

Technology Needed: No

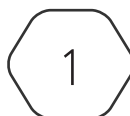
Approximate Time Required: 45-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.1 - Explain fundamental security concepts and frameworks.
 - **Authentication, authorization, accounting, and non-repudiation concepts**
 - Authentication
 - Single factor
 - Multifactor
 - Single sign-on
 - Authorization
 - Permissions
 - Administrator vs. user accounts
 - Least privilege model
 - Accounting
 - Logs
 - Location tracking
 - Web browser history

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Lesson Background

Background Information:

In the world of cybersecurity, trust is not enough – we need proof. When people use digital systems, they may try to deny their actions later, whether it's sending a message, downloading a file, or changing important data. This is where non-repudiation comes in.

Non-repudiation ensures that a user cannot deny taking a specific action within a system. It provides verifiable evidence that proves who did what, when, and where. This is especially important in situations like online banking, email communication, or school systems where accountability matters.

Non-repudiation ties directly into the AAA triad: Authentication verifies who the user is. Authorization verifies what the user is allowed to do. Accounting records when and how actions were taken.

Together, these create a digital trail that ensures people are held responsible for their actions. Without non-repudiation, trust in digital systems would collapse – anyone could deny their actions without proof. By understanding and analyzing non-repudiation, students will learn how accountability is maintained in technology systems and why it is a critical part of security.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.1.8 – Non-Repudiation • Activity 6.1.8 – We Can Prove It
Materials per Student	• Student Notes 6.1.8 – Non-Repudiation • Student Handout 6.1.8 – Non-Repudiation • Assessment 6.1.8 – Non-Repudiation (optional)

Cyber Terms:

Term	Definition
AAA Triad	a cybersecurity framework that secures access to systems and data by making sure users are who they say they are (Authentication), can only do what they're allowed to do (Authorization), and that their actions are recorded (Accounting)
non-repudiation	ensures that a user cannot deny having performed an action or transaction

6.1.8 – Non-Repudiation

Guiding Question: How can organizations ensure that users cannot deny their actions in a digital system?

Lesson Launch (10-15 minutes)

- **Ask:** “Someone posted a comment online and later claims they never wrote it. How could someone prove it was really them?”
- **Say:** “In cybersecurity, **non-repudiation** is a concept that ensures that people cannot deny their actions when using a computer system.”
- Just like signing a receipt at the store proves you bought something or sending a tracked email proves you sent the message, non-repudiation proves your actions on a digital system.
- Remind students of the **AAA Triad** and its relation to non-repudiation.
 - Explain that non-repudiation depends on all three parts of the AAA Triad. If we know who did it, what they were allowed to do, and we have a record of their action, they cannot claim they didn’t do it.
 - Authentication: You log in with your username and password to prove your identity.
 - Authorization: Once logged in, you can open your assignments but not change your grades.
 - Accounting: The system records when you log in and what files or resources you accessed.
 - **Non-repudiation: If you submit an assignment, you can’t deny it later because your account and timestamp prove it was you.**
- A bank can show a logged record proving a customer transferred money.
- A gaming platform can show who made an in-game purchase, preventing disputes.
- **Ask:** Why do you think these logs and proofs matter?

No Denying It (10-15 minutes)

- Distribute the Student Handout 6.1.8 – Non-Repudiation and have students fill in the answers during the lesson.
- Discuss non-repudiation, its definition, foundation, benefits, and examples.
 - It is crucial for accountability, legal compliance, and security in systems like banking, email, or corporate networks.
 - **Non-repudiation** is like a digital receipt that proves responsibility.
 - Benefits of non-repudiation:
 - Accountability: People are responsible for their actions.
 - Trust: Organizations can trust digital communications and transactions.

- Legal Protection: In court, non-repudiation is proof of who did what.
- Security: Prevents fraud, disputes, and 'blame-shifting'

Activity: We Can Prove It (20-25 minutes)

- Display Activity Slides 6.1.8 – We Can Prove It, have students record their answers in the handout.
- Decide if students will work independently or in small groups.
- Read each scenario aloud (or allow students/groups to read independently).
- Students will determine how the AAA Triad aids non-repudiation when people try to deny their actions online.
- Allow time for students to write their answers.

Scenarios and Answers

1. Art claims he did not submit an online quiz, but the teacher sees it marked as submitted. Who likely submitted the quiz? What proof would confirm it? How does the AAA Triad help?

Art likely submitted the quiz because submission is marked on his account. Proof could be the submission timestamp, account login info, and the IP address. Authentication confirms the identity of the user, authorization confirms the user had permission to submit, and accounting provides a timestamped record for non-repudiation.

2. A banking system shows \$50 transferred from Rainbow's account. Rainbow claims she didn't make the transfer. How can the bank verify who performed the transfer? What evidence supports non-repudiation? What if someone got Rainbow's account info?

The bank can check login history, device IP address, and transaction logs. Authentication confirms which user was logged in, authorization confirms if Rainbow's account had permission for transfers, and accounting shows the timestamp, amount, and target account. If someone got Rainbow's account info, non-repudiation might fail, and MFA could strengthen authentication proof.

3. A shared class document has edits that were not approved by the group. Taylor and Jamie both claim they didn't make the edits. How can the group determine who edited the document? What information from the system could prevent denial? What if logins or shared devices overlap?

The group can check edit history, usernames, timestamps, and device IP addresses. Authentication shows which user made the edits, authorization confirms if the user was allowed to edit, and accounting provides proof of who did what and when. If logins or shared devices overlap, non-repudiation will be more complicated.

4. A company email with sensitive info was sent to the wrong recipient. Cooper claims he didn't send it. How can IT confirm who sent the email? How can non-repudiation prevent disputes?

The company should examine email logs viewing the sender account, device, timestamp, and IP address. Authentication confirms the user's identity, authorization confirms the sender had permission to send company emails, and accounting logs provide a traceable record showing the action occurred. Because of the AAA Triad, non-repudiation will prevent Cooper from denying he sent the email if the evidence shows it is him.

5. Georgia claims she didn't delete a group chat message, but it disappeared. What evidence could confirm who deleted the message? How does the AAA Triad apply in this case? What limits might make non-repudiation harder here?

Evidence would consist of chat logs, login sessions, and device info. Authentication confirms which account was active, authorization shows that only members with delete permissions could remove messages, accounting logs show timestamps and devices. If accounts are shared, non-repudiation may be weaker.

6. A confidential file was downloaded from a shared folder. Jess denies downloading it. How can the system prove who downloaded the file? What logs would strengthen non-repudiation here? What risks exist if users share accounts?

Download logs, usernames, and device/IP addresses can be reviewed. Authentication confirms which user was logged in, authorization confirms permission to access the file, and accounting logs show timestamp and location. Shared accounts make non-repudiation harder and reduce accountability.

7. A restricted app was installed on a company laptop. Monique claims she didn't install it. How can IT confirm which user installed the app? Which AAA element is most important here? What controls could make this system more tamper-proof?

Installation logs, login sessions, and device info can confirm who installed the app. Authentication shows which user account was active, authorization restricts installation rights, and accounting logs confirm the exact time of the installation. Stronger permissions and MFA would strengthen accountability.

8. Camber's grade was changed in the online gradebook. She denies making the change. How could logs disprove Camber's claim? Why is non-repudiation important in grade systems? What security measures can prevent unauthorized grade changes?

Logs track edits, timestamps, and user accounts. Authentication shows who was logged in, authorization ensures only teachers or principals can change grades, and accounting confirms who did what and when. Stronger permissions could prevent unauthorized grade changes.

9. A message containing private info was accidentally posted to social media. Isaiah claims he didn't post it. What records could prove who posted the message? Which AAA element protects against false denial here? What if Isaiah's device was left unlocked?

Logs can show the poster's account, timestamps, device, and IP address. Authentication confirms account identity, authorization confirms posting permissions, and accounting proves when and where the info was posted from. If Isaiah's device was left unlocked it is a weakness for non-repudiation because anyone could have used his phone to make the post.

10. Someone accessed a blocked website on the school network. Nick denies doing it. What logs could confirm who accessed the site? How do logs help provide non-repudiation? What challenges exist if multiple students share a device?

Logs record the URL, user credentials, and device/IP address. Authentication confirms which student was logged in, authorization shows blocked vs. allowed site rules, and accounting confirms exact actions with timestamps. Shared devices complicate proof of responsibility and weaken non-repudiation.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What is non-repudiation, and how does it build on the AAA Triad?

Non-repudiation means users cannot deny their actions. It builds on authentication (proving identity), authorization (checking permissions), and accounting (recording actions) by making sure those records stand as proof later.

- What could happen without non-repudiation, and why would that be a problem?

Without it, people could deny harmful or accidental actions like sending a message, moving money, or changing grades. This creates mistrust and makes it hard to hold people accountable.

- Why is non-repudiation especially important in digital systems compared to face-to-face interactions?

In person, people can often see who did something. Online, identity is harder to prove, so non-repudiation tools provide the “digital proof” we need for trust.